Measurement Systems

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Measurement Systems (A09892)

Short Title:	Measurement Systems		
Faculty	Science and Computing		
Department	Science		
Cluster	Engineering		
Author	CWALSH		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

The emphasis of the module will be towards design, analysis and implementation of a number of complete measurement systems. There will be a comprehensive study of the general principles of measurement systems, including sensors, signal conditioning, signal processing and presentation. A number of specialised systems will be developed in the laboratory programme of this module, using a graphical programming language (National Instruments LabVIEW).

Programmes

`COMP-0411`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	3 / 5 / E
`COMP-0411`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	3 / 5 / M
	BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	3 / 5 / M
`COMP-0411`	BSc (Hons) in the Internet of Things (`WD_KINTT_B`)	3 / 5 / M

Indicative Content

- General principles: general measurement systems; characteristics of measurement system elements
- Signals and noise in measurement systems: effects of noise on measurement circuits; methods of reducing the effects of noise
- Sensing elements: resistive, capacitive, inductive, electromagnetic, thermoelectric and ultrasonic
- Signal conditioning: potential divider, bridge circuits; general amplifier characteristics, operational amplifiers, feedback amplifiers
- Signal processing: active filters; low-pass, high-pass, band-pass and band-stop; analog to digital interconversion
- Sensor science and the study of a number of specialised sensor systems

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Examine the general principles and characteristics of measurement systems.
2. Examine the various types of noise and demonstrate methods of reducing the effects of noise in measurement circuits.
3. Examine the various signal conditioning elements of a measurement system.
4. Examine the principles of various signal processing elements of a measurement system.
5. Appraise the critical scientific and technological characteristics and applicability of a broad range of sensor systems.
6. Develop experimental skills in the design, implementation, testing and reporting of various elements of a measurement system.
7. Design and test experiments using National Instruments Multisim and LabVIEW.
8. Prepare and deliver a presentation to disseminate results of practical work.

Learning and Teaching Methods

- Lectures: Lectures will be used to introduce new topics and their related concepts.
- Practicals: The practical element will allow students to design implement and test systems, interpret and report experimental results.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Presentation	10%	8
Practical	30%	6,7

Assessment Criteria

<40%: Unable to demonstrate a basic understanding of elementary measurement systems, sensors and sensor technology. Experimental work is generally poor.

40%-49%: Able to provide partial solutions for complete systems. Display limited knowledge of the principles of measurement systems and the science of sensors. Practical work reaches an acceptable level.

50%-59%: Able to describe clearly principles of operation of a range of sensors. Able to analyse specialised measurement systems. Practical work is performed and reported on in a capable manner.

60%-69%: Demonstrates an ability to design a specialised measurement system from a number of component parts. Has a high level of experimental skills and has shown the ability to produce good, clear reporting of practical work.

70%-100%: All the previous to an excellent level. Demonstrates an understanding of the limitations of circuits and measurement systems and subsequent consequences. Assess whether solutions are meaningful. Can critically assess any experimental work undertaken.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	24	
Independent Learning	75	

Supplementary Material

- Bentley, J. _Principles of Measurement Systems_. 4th ed. UK: Prentice-Hall, 2008.
- Doebelin, E. _Measurement Systems Applications and Design_. 5th ed. United States of America: McGraw-Hill, 2003.

- Horowitz, P. and W. Hill. _The Art of Electronics_. 3rd ed. UK: Cambridge University Press, 2015.
- Waldemar, N. _Measurement Systems and Sensors_. 2nd ed. United States of America: Artech House, 2015.

Requested Resources

- SCIENCE LAB: Physics